

Flyback SMPS Design Session – Detailed Engineering Notes

Project Overview

This document summarizes the complete design, validation, and measurement process of a custom offline flyback SMPS. The project demonstrates end-to-end power electronics engineering, including hand-wound magnetics, control-loop design, drain stress validation, noise mitigation, and efficiency characterization, suitable for professional portfolio and job applications.

System Specifications

Topology: Isolated Flyback Converter

Input: 120 VAC (rectified to ~ 170 VDC bus)

Prototype Output: 5 V @ 0.3 A

Planned PCB Output: 5 V @ 1 A

Controller: TinySwitch (TNY285-class)

Feedback: TL431 + optocoupler

Isolation: Transformer + optocoupler (no auxiliary winding)

Primary-Side Architecture

The TinySwitch controller operates without an auxiliary winding, relying on internal drain-derived biasing. The EN/UV pin is populated with a high-value resistor (~ 2.7 M Ω) from the rectified HV bus. Regulation is achieved by the optocoupler transistor pulling EN/UV low in response to TL431 feedback.

Custom High-Frequency Transformer

A fully custom, hand-wound HF transformer was designed without reference magnetics. Turns ratio, magnetizing inductance, and leakage inductance were validated indirectly through stable regulation, controlled drain stress, and acceptable reflected voltage behavior.

Primary RCD Clamp Validation

The RCD clamp limits drain overvoltage caused by leakage inductance. Drain-to-source voltage was measured using a 500 $\times$ differential probe. At 120 VAC input, the measured peak VDS was approximately 330–340 V, providing substantial margin to the MOSFET rating.

Secondary-Side Power Path

The secondary uses a Schottky rectifier appropriate for low-voltage output. Initial output capacitance was minimal, leading to measurable ripple. The design was refined to include ~ 1000 μ F total electrolytic capacitance and plans for additional ceramic filtering.

Feedback Loop Design

The TL431 provides error amplification with a resistor divider setting 5.00 V output. Divider dissipation was calculated in the milliwatt range, making 1/8 W or 1/4 W resistors sufficient. A small compensation capacitor (~ 1 nF) between TL431 cathode and REF ensured loop stability.

Load Regulation Testing

Load testing was performed at 50 mA, 100 mA, 227 mA, and approximately 300 mA. The output remained tightly regulated at 5.00 V with no audible noise or instability, confirming correct loop gain and transformer design.

Drain Waveform Analysis

Observed drain waveforms showed burst-mode operation at light load and continuous switching at higher load. Leakage ringing was present but well damped. Absolute $V_{DS,max}$ was determined by DC-coupled measurements, not peak-to-peak readings, ensuring correct stress evaluation.

Output Noise Characterization

Two noise components were identified: high-frequency spikes from diode recovery and leakage, and low-frequency ripple from energy packet delivery. Mitigation includes a secondary RC snubber (~ 3.3 nF + 3.3 Ω), increased bulk capacitance, and addition of μ F-range X7R ceramic capacitors at the rectifier node.

Efficiency Measurement Strategy

True efficiency is defined from AC wall input to DC output. Preferred measurement uses real AC input power via wattmeter or scope-based $v(t)i(t)$ integration. DC bus power measurement is also used to characterize converter-stage efficiency separately.

USB Charger Adaptation

To function as a phone charger, the design requires a USB connector and proper port identification. USB-A may use BC1.2 DCP signaling (shorted D+/D-). USB-C requires 5.1 k Ω pull-down resistors on CC1 and CC2 to advertise default power.

Middlebrook Loop Injection Plan

For the TinySwitch design, injection is performed in the optocoupler-to-EN/UV path using a transformer and DC keep-alive resistor. Sweeps from 10 Hz to 50 kHz allow extraction of crossover frequency and phase margin. Future UCC designs will use the COMP pin for cleaner injection.

Future 20 V / 5 A UCC Flyback

The next project targets ~ 100 W output. This significantly increases magnetics, EMI, and thermal challenges. A UCC-class controller with a COMP pin simplifies loop analysis and supports higher power operation.

Professional Assessment

This project demonstrates complete offline SMPS design competence: magnetics, control loops, protection, measurement, and documentation. The work is directly comparable to entry- to mid-level power electronics engineering tasks.

Suggested Resume Framing

Designed and hand-wound an isolated offline flyback SMPS. Implemented TL431/opto feedback, validated drain stress and clamp behavior, characterized efficiency vs load, and mitigated output noise via snubber and filtering. Prepared design for PCB integration and loop-gain measurement.